

Exponential Equations and an Introduction to Logarithms (Part II) Homework

Solve. Write the Roman numeral of each exponent rule that you use.

① $81^{2x-5} = 27$ ② $\frac{1}{16^x} = 64^{5-x}$ ③ $\sqrt[3]{4} = 32^{6x}$

④ $36^{x+3} = 216^{-4x+4}$

Solve. Write the Roman numeral of each exponent rule that you use. If an answer is irrational, round to the nearest ten thousandth.

⑤ $3 - 2(3^{8x+3}) = 1$ ⑥ $1 = -9(4^{1+3x}) + 37$

⑦ $\frac{6^x}{2} + 4 = 5$ ⑧ $26 = 5 + 3(2^{7x-6})$

⑨ $1 = \frac{3^{2-2x}}{4} - 6$ ⑩ $17 = 2(5^{\frac{x}{3}+4}) - 9$

⑪ $8 + 5(7^{3-\frac{x}{2}}) = 10$ ⑫ $70 = 5(2^{x^2-2x}) + 30$

⑬ $4 - 2(3^{-x^2-5x}) = -158$ ⑭ $\frac{1}{1,000} = 10^{x^2} \cdot 10^{4x}$

⑮ $\frac{5^{x^2}}{5^x} = 25$ ⑯ $4^{2x} - 9(4^x) + 20 = 0$

⑰ $0 = 2(5^{2x}) - 17(5^x) - 9$

Solve. Write the Roman numeral of each exponent rule that you use. If a logarithm is irrational, round to the nearest ten thousandth.

⑱ $9^{x^2} \cdot 9^{6x} = 81$ ⑲ $-2(7^{2x^2+3x}) + 4 = -10$

⑳ $11^{2x} + 12(11^x) + 20 = 0$ ㉑ $0 = 6(9^{2x}) + 23(9^x) - 13$

Answers

① $x = \frac{2^3}{8}$

⑫ $x = 3 \text{ or } x = -1$

② $x = 15$

⑬ $x = -1 \text{ or } x = -4$

③ $x = \frac{1}{45}$

⑭ $x = -3 \text{ or } x = -1$

④ $x = \frac{3}{7}$

⑮ $x = -1 \text{ or } x = 2$

⑤ $x = -\frac{3}{8}$

⑯ $x = 1 \text{ or } x \approx 1.161$

⑥ $x = 0$

⑰ $x \approx 1.3652$

⑦ $x \approx 0.3869$

⑱ $x = -3 \pm \sqrt{11}$

⑧ $x \approx 1.2582$

⑲ $x = \frac{-3 \pm \sqrt{17}}{4}$

⑨ $x \approx -0.5166$

⑳ No Solution

⑩ $x \approx -7.2189$

㉑ $x \approx -0.3155$

⑪ $x \approx 6.9418$